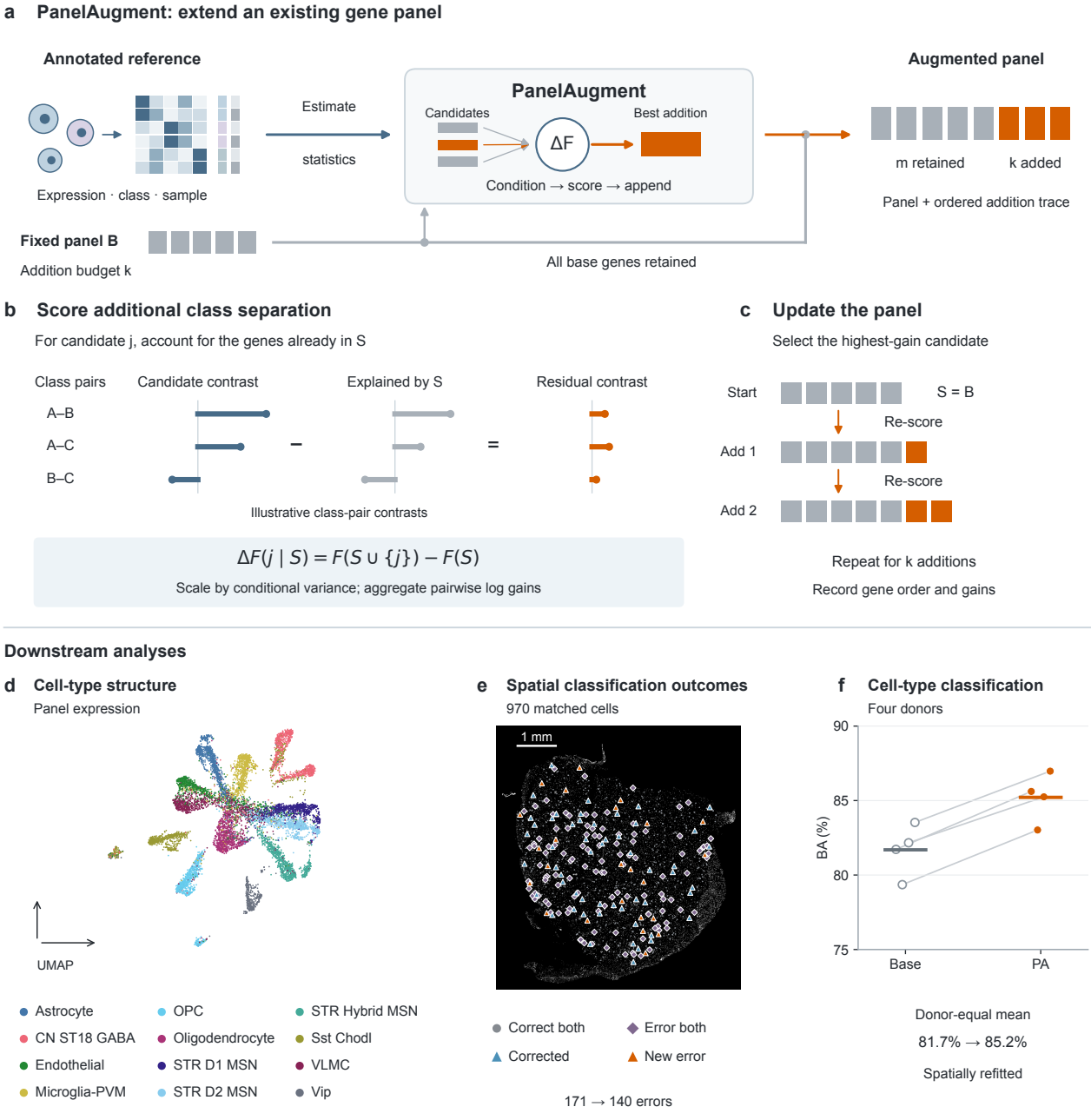

a
PanelAugment: extend an existing gene panel
Annotated reference
Expression · class · sample
Estimate
statistics
PanelAugment
Candidates
ΔF
Best addition
Condition → score → append
Augmented panel
m retained
k added
Panel + ordered addition trace
Fixed panel B
Addition budget k
All base genes retained
b
Score additional class separation
For candidate j, account for the genes already in S
Class pairs
Candidate contrast
Explained by S
Residual contrast
A–B
A–C
B–C
Illustrative class-pair contrasts
$\Delta F(j \mid S) = F(S \cup \{j\}) - F(S)$
Scale by conditional variance; aggregate pairwise log gains
c
Update the panel
Select the highest-gain candidate
Start
S = B
Re-score
Add 1
Re-score
Add 2
Repeat for k additions
Record gene order and gains
Downstream analyses
d
Cell-type structure
Panel expression
UMAP
Astrocyte
CN ST18 GABA
Endothelial
Microglia-PVM
OPC
Oligodendrocyte
STR D1 MSN
STR D2 MSN
STR Hybrid MSN
Sst Chodl
VLMC
Vip
e
Spatial classification outcomes
970 matched cells
1 mm
Correct both
Error both
Corrected
New error
171 → 140 errors
f
Cell-type classification
Four donors
BA (%)
90
85
80
75
Base
PA
Donor-equal mean
81.7% → 85.2%
Spatially refitted